

Perovskite Photovoltaics in June 2026: Defect passivation, composition and phase control

Havid Aqoma^{1,2,3,4,*}

¹School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

²Kelip-kelip! Center of Excellence for Light Enabling Technologies, School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

³Centre of Excellence for Green and Advanced Technologies in Efficient Separations (GATES), School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

⁴Gulei Innovation Institute, Xiamen University, Zhangzhou 363200, China

*Corresponding author. Email: havidaqoma.khoiruddin@xmu.edu.my

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Abstract

From a monthly corpus of 623 indexed publications in perovskite photovoltaics, critical examination of 167 works reveals that unstandardized characterization continues to conflate reversible ionic polarization with enduring chemical defect remediation. Only 14 studies report independently certified performance metrics, establishing a verified efficiency boundary of 33.6% for tandem configurations and 27.28% for single-junction devices. These certified frontiers rely on multidentate Lewis-base chelators that elevate vacancy formation energies, dense self-assembled monolayer (SAM) hole contacts that suppress interfacial deprotonation, and interfacial strain relief that impedes nucleation of hexagonal non-perovskite phases. However, severe geometric scaling penalties persist across larger formats, reducing certified power conversion efficiency (PCE) to 26.8% at an aperture area of 655.2 cm² as macroscopic contact pinholes, sheet resistance, and solvent washouts induce non-radiative recombination. Standardized operational stability evidence remains thin: merely 6 investigations document operational lifetime thresholds, with the longest duration reaching 3600 h, and only 3 adhere to International Summit on Organic Photovoltaic Stability (ISOS) testing procedures. A pervasive reporting disparity accentuates this ambiguity, as 48.1% of publications declare efficiency values while only 5.9% provide independent certification, leaving uncertified claims vulnerable to sweep-rate hysteresis, aperture misalignment, and capacitive transients. Mechanistically, evidence confirms that multidentate coordination and steric hindrance mitigate lead vacancy formation and fullerene photo-dimerization, while catalytic anchoring protects selective contacts against solvent erosion. Nevertheless, macroscopic work-function shifts routinely conceal severe nanoscale contact disorder and trapped-hole accumulation. Bulk structural probes fail to distinguish thermodynamic phase stabilization from kinetically trapped metastable states, and macroscopic measurements conceal substantial morphological disparities across sequentially deposited scaffolds. Because current lifetime testing lacks uniform electrical bias and thermal baselines, available data cannot verify whether contact reconstruction confers greater operational longevity than molecular passivation, necessitating operando impedance spectroscopy and drive-level capacitance profiling under continuous stress.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Standard steady-state characterization cannot distinguish reversible ionic redistribution from irreversible chemical decomposition, leaving ambiguous whether reported passivation strategies eliminate recombination centres or merely establish transient screening fields. Resolving this ambiguity defines the central challenge in perovskite photovoltaics, where the research frontier has departed from raw power conversion efficiency (PCE) recorded on small-area coupons. Progress now requires reproducible independent verification, structural uniformity over scalable device areas, and operating lifetime that resists coupled thermal, environmental, and electrical stress.

A total of 623 indexed works met the scope gate in June 2026 and 167 were read closely. Across this evidence base, this review distinguishes what the month’s literature verified through corroborated physical measurement from what it asserted through uncalibrated fits, isolated champion cells, or non-standard aging protocols.

Five mechanistic threads track the findings that withstand that measurement uncertainty: non-radiative recombination and defect control, crystallisation pathways and phase stability, interfacial energetics and contact design, operational lifetime and failure pathways, and subcell design and tandem interconnection. Tracing these processes clarifies where microscopic defect passivation translates into authentic device endurance, and where nominal performance gains remain artifacts of measurement conditions.

2. The Certified Frontier in June 2026

Unstandardized laboratory current-voltage measurements cannot distinguish interfacial defect passivation from transient capacitive currents, sweep-rate hysteresis, spectral mismatch, or aperture misalignment. Only 14 of 167 closely read papers reported an independently certified efficiency, leaving the majority of performance claims unshielded from these experimental ambiguities. What survives standardization establishes the verified photovoltaic ceiling. Two-terminal (2T) tandems set the highest certified PCE at 33.6% and 32.95%, with a lower certified baseline at 28.79% [1,2,3]. Unclassified cell architectures reach certified efficiencies of 27.28% and 26.84% [4,5], while certified inverted (p-i-n) devices cap at 26.8% and 26.44% [6,7].

For scalable modules, the certified ceiling drops to 26.61% [8]. Self-reported values at large aperture areas run substantially ahead of this verified module boundary, reaching 27.12% on 655.2 cm² [6], 26.88% on 29.7 cm² [9], 26.76% on 12 cm² [10], 26.52% on 50 cm² [11], and 26.13% on 57.6 cm² [2]. A champion efficiency recorded on a tiny laboratory cell and an independently certified efficiency measured at module area provide fundamentally distinct physical evidence. The miniature laboratory cell interrogates local non-radiative recombination suppression under negligible series resistance and idealized collection geometries. By contrast, a certified module reflects distributed transport physics, penalizing performance through interconnect dead areas, sheet resistance across transparent conductors, and film thickness gradients over large spans. Treating micro-scale champion metrics as indicators of module feasibility conflates intrinsic material quality with macroscopic charge-transport management.

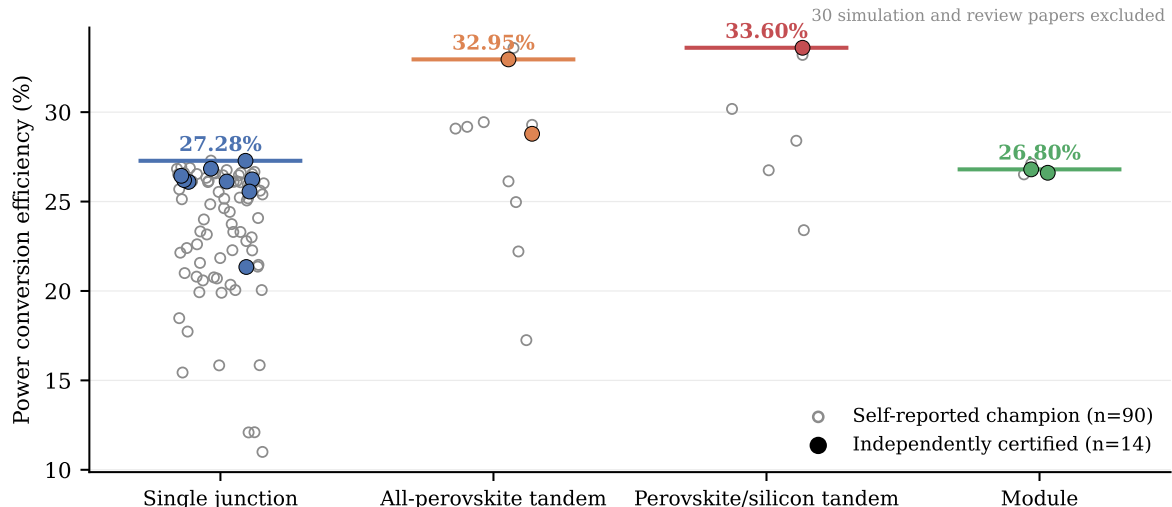


Figure 1. The certified frontier in June 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Non-Radiative Recombination and Defect Control

Standard optoelectronic characterization cannot definitively separate true chemical passivation of deep sub-gap trap states from electrostatic band-bending, interfacial dipole reorientation, or suppressed mobile ion accumulation that dynamically alters interface recombination. Most reported open-circuit voltage (V_{oc}) and photoluminescence improvements conflate these effects. Evidence of genuine chemical trap remediation survives where multidentate Lewis-base chelators target undercoordinated lead (Pb^{2+}) cations and halide vacancies. Bidentate coordination directly elevates defect formation energies and prevents ligand detachment: dabrafenib (DRFN) binds multiple Pb^{2+} centers through its sulfonyl group and heterocyclic nitrogen to yield uncertified power conversion efficiencies (PCE) of 25.28% in n-i-p and 26.62% in p-i-n cells [12], while dithiol chelation (DPS) forms stable five-membered rings with Pb^{2+} to achieve 22.28% PCE [13]. Enantiomeric anchoring reveals that steric configuration dictates coordination strength: chiral R-Pye raises the iodine vacancy formation energy from 2.46 to 4.56 eV with a 9.45 eV binding energy compared to 7.33 eV for S-Pye, delivering 26.13% PCE [14]. Similarly, functionalized aromatics like 2-amino-4,6-dichlorotriazine relieve PbI_6 octahedral distortion while coordinating Pb^{2+} and halide vacancies (26.58% PCE) [15], paralleled by triazine anchoring that suppresses Pb-I antisites [16], amino-pyridine derivatives (ACA) strengthening Pb^{2+} coordination via electron donation while hydrogen-bonding lattice iodide (15.84% PCE) [17], and dual-site binding through carbonyl, phenolic, or cyano groups [18,19,20].

This mechanistic picture is obscured when molecular treatments simultaneously alter bulk crystal growth, leaving unclear whether non-radiative recombination decreases from localized defect passivation or reduced grain boundary area. The dual-functional ionic liquid PIT decouples these actions: its anion coordinates Pb^{2+} and organic cations, whereas its cation segregates to the upper surface to retard crystallization and enlarge grains, achieving 25.94% PCE [21]. Co-passivation with potassium diformate and potassium phosphate similarly halves trap density while expanding grain size from 780 to 1370 nm (24.42% PCE) [22]. At buried contacts, separating chemical passivation from nucleation control remains equally challenging. Adding $CsPbBr_3$ nanocrystal seeds passivates the buried

interface while steering crystallization, yielding 24.63% PCE against 22.89% for unseeded controls [23]. Interfacial polymeric interlayers like ultraviolet (UV)-crosslinked RM82 diacrylates passivate buried tin oxide (SnO_2) interfaces while promoting crystal growth, though reported stability relies on passive ambient storage rather than operational stress [24]. CTW-g-PANI coordinates undercoordinated tin sites to heal SnO_x oxygen vacancies and passivate surface perovskite states (17.73% PCE) [25]. In tin-lead perovskites, differential chelation resolves mismatched crystallization kinetics: 1,10-phenanthroline binds Sn^{2+} preferentially over Pb^{2+} , synchronizing film growth to achieve an uncertified 23.63% PCE, 81.27% fill factor (FF), and 0.887 V V_{oc} [26].

Static defect metrics also cannot distinguish permanent electronic trap deactivation from temporary trap filling or the suppression of field-driven ion migration. Although illumination-assisted trap filling achieves long electron lifetimes up to $547 \pm 9 \mu\text{s}$ in polycrystalline films [27], unpassivated mobile ions migrate under operational bias to generate new recombination centers. Defect suppression only survives long-term operation when ion migration pathways are physically sealed. 2PyS coordination suppresses undercoordinated Pb^{2+} and halide vacancies, limiting ion drift and preserving over 91% efficiency after 2000 h in wide-bandgap (WBG) cells [28], consistent with molecular confinement domains restricting formamidinium motion to arrest halide segregation (24.21% and 21.20% PCE for 1.68 and 1.78 eV bandgaps) [29]. Time-of-flight secondary ion mass spectrometry (ToF-SIMS) confirms that multidentate ATZCl forms a Ruddlesden-Popper (RP)-like surface phase that blocks iodide migration across the hole transport layer (hole-transport layer (HTL)) (25.22% PCE) [30]. Replacing small molecules with in situ polymerized vinylphosphonic acid provides a rare certified 26.24% PCE (26.54% uncertified champion) with over 90% retention after 1600 h of continuous maximum power point tracking (MPPT) [31], demonstrating that covalent network immobilization prevents field-induced passivator detachment and settles the ambiguity between chemical trap deactivation and dynamic ion blocking.

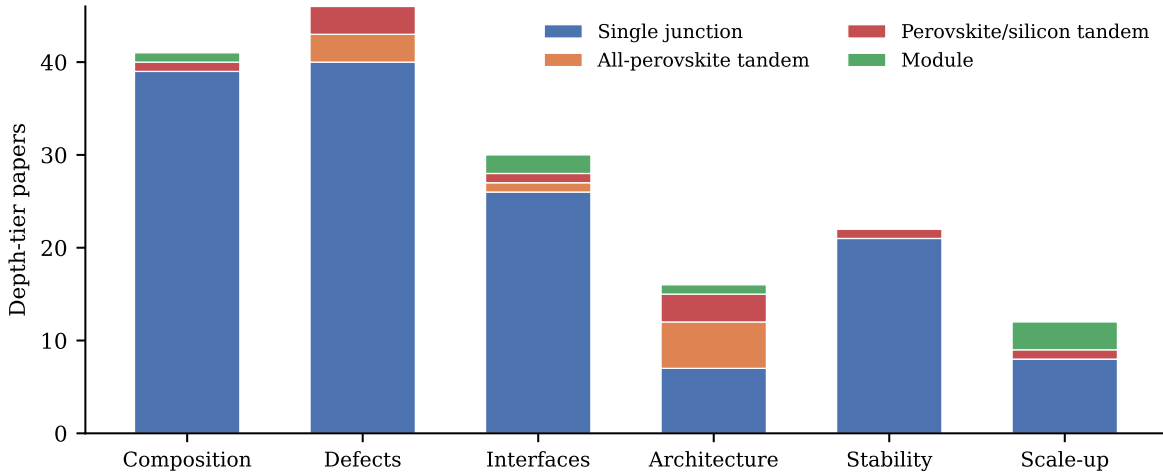


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Crystallisation Pathways and Phase Stability

Bulk structural probes cannot distinguish whether additives genuinely alter the thermodynamic free-energy landscape of perovskite phases or merely impose kinetic barriers that freeze metastable

states [32]. Spatially averaged X-ray diffraction (XRD) routinely masks localized phase impurities, while thermal calibration offsets skew the determination of true phase-transition and decomposition onsets in halide powders [32,33]. What survives this diagnostic ambiguity is that local interfacial lattice matching and coherent strain govern polymorphic stability. In formamidinium lead triiodide (FAPbI₃), the spontaneous conversion from the photoactive black cubic perovskite (α -phase) to the non-photoactive yellow hexagonal (β -phase) at room temperature nucleates at coherent interfaces between α -phase (011) and β -phase (110) planes under tensile strain [34,35]. Introducing two-dimensional (2D) perovskitoid interlayers disrupts this preferred α -phase orientation, yielding an uncertified 25.61% champion PCE [34]. Interfacial strain regulation is equally achieved by constructing one-dimensional (1D) THAPbI₃ heterojunctions using tetrahexylammonium iodide (THAI), which anchors formamidinium (FA⁺) cations through hydrogen bonding to suppress outward cation diffusion and strain gradients, achieving an uncertified 25.06% PCE on unstated cell dimensions [36]. Transient intermediate phases also dictate phase-pure crystallization pathways. Volatile thiocyanate (SCN⁻) anions transiently coordinate with lead (Pb²⁺) precursors to restructure crystallization kinetics; while ammonium (NH₄⁺) cations volatilize during thermal annealing to drive bulk grain enlargement, residual phenylethylammonium (PEA⁺) cations segregate to grain boundaries to provide surface passivation, achieving 23.3% PCE [37]. In all-inorganic cesium lead triiodide (CsPbI₃), formamidinium acetate (FAAC) reduces the formation energy of the Cs₄PbI₆ intermediate phase, facilitating complete dimethylammonium (DMA⁺) to cesium (Cs⁺) cation exchange and accelerating dimethylammonium iodide (DMAI) removal to yield 21.84% PCE [38].

In wide-bandgap mixed-halide absorbers, measurements similarly struggle to distinguish growth-induced compositional heterogeneity from operational light-induced phase segregation [32,39]. Resolving this requires synchronizing disparate halide crystallization kinetics. Eutectic urea phosphate, possessing stronger affinity for bromine-based over iodine-based octahedra, equalizes reaction rates to eliminate vertical and lateral phase segregation, yielding 22.27% PCE [40]. Epitaxial growth on spinel oxide substrates uses lattice matching and magnesium-halide interfacial bonds to induce compressive strain, which arrests halide segregation without organic additives [41]. For multi-junction absorbers, substituting bromide with chloride in Cs_{0.05}(FA_{0.90}MA_{0.10})_{0.95}Pb(I_{0.95}Br_{0.05})₃ (incorporating methylammonium, MA⁺) circumvents performance losses observed beyond 450 nm thickness, enabling crack-free vertical crystallization up to $\sim$ 700 nm without sacrificing FF or open-circuit voltage (Voc) to achieve 23.4% PCE on a 1 cm² aperture [42].

Controlling orientation templates is critical for dimensional stability. Templated crystallization using pre-synthesized large-area $n = 1$ 2D perovskite nanosheets drives three-dimensional (3D) film growth with high crystallinity and preferred orientation, achieving 21.36% PCE (20.75% stabilized power output) and a hysteresis coefficient below 1% [43]. Propylammonium iodide (PAI) added during two-step blade coating directs (111)-oriented growth in inverted cells, delivering 20.05% PCE and retaining 95% of initial efficiency over 1680 h of MPPT [44]. Conversely, solvent-additive approaches such as rubidium chloride (RbCl) in 2-methoxyethanol and dimethylacetamide (DMAc) claim methylammonium lead triiodide (MAPbI₃) phase regulation, yet stability verification is limited to 300 s of steady-state output at 18.48% PCE [45]. In quasi-2D systems, functional group configuration in isomeric additives dictates molecular stacking: cytosine exhibits stronger delocalized interactions than iso-cytosine, facilitating favorable nucleation to attain 22.4% PCE with a T80 lifetime (time to 80% initial efficiency) of 3600 h [46]. Intermolecular pi-pi interactions in 3,3'-methylenediphenyldiammonium cations stabilize a non-Dion-Jacobson (non-DJ) 2D perovskite, yielding 26.52% module PCE on 50 cm² [11]. For tin-based perovskites, density functional theory (DFT)

guided screening of solvent vapor fuming identifies tetramethylurea (TMU), which slows coordination kinetics to enable carrier extraction in 0.67 s and a 75.14 s recombination lifetime, reaching 10.45% PCE [47].

5. Interfacial Energetics and Contact Design

Macroscopic junction characterizations fail to distinguish whether contact modifications enhance carrier collection through electrostatic band alignment or through the physical elimination of local pinholes, solvent washouts, and interfacial deprotonation. Transient photovoltage and photocurrent mapping reveals that inhomogeneous hole transport layer (HTL) junctions generate trapped-hole densities up to $1.6 \times 10^{17} \text{ cm}^{-3}$ [48], confirming that macroscopic work function shifts routinely conceal severe nanoscale contact disorder. Under operational stress, unimolecular self-assembled monolayer (SAM) contacts undergo severe degradation driven by thermomechanical mismatch with the perovskite absorber [49]. What survives this diagnostic ambiguity is direct evidence that contact durability depends on molecular ordering, multi-site anchoring, and chemical buffering against proton transfer rather than energetic tuning alone.

Because casting perovskite precursor solutions washes away unanchored molecules, contact integrity requires accelerated chemisorption or continuous replenishing. Catalytic deposition using 4-dimethylaminopyridine (DMAP) accelerates phosphonic acid binding to indium tin oxide (ITO) via phosphorylpyridinium intermediates, achieving a certified PCE of 26.44% (champion 26.93%) [7], while directly doping SAM molecules into precursor solutions or inserting alumina nanoparticles counters solvent erosion and seals pinholes [50,51]. At the substrate interface, uniform coverage requires suppressing molecular agglomeration; bisphosphonic anchors, seed layers, and rigidified conjugated backbones enforce packing uniformity and favorable dipole alignment without excessive islanding [52,53,54,55]. Specifically, rigid phenyl linkers elevate champion PCE to 23.30% compared to 22.38% for flexible alkyl analogues [54], and nonplanar butterfly geometries maintain 97.5% efficiency after 1000 h under International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-1 testing [55]. To resist interfacial shearing, multidentate and polymeric architectures buffer stress: a multidentate phosphonic acid network boosts fracture toughness nearly eightfold to 9.11 MPa (certified 26.61% PCE, though on an unscaled 0.06734 cm^2 area) [8], whereas polymeric Poly-3Ph-4PACz delivers 22.22% PCE on 57.6 cm^2 flexible modules [2].

At electron transport layer (electron-transport layer (ETL)) contacts, performance gains are frequently misattributed to work function realignment when chemical degradation dominates. Basic hydroxyls on tin oxide (SnO_2) promote formamidinium (FA^+) deprotonation, a failure mode arrested by phosphate buffering or hydrogen-bonding additives at buried junctions [56,57]. Concurrently, bilateral functional molecules mitigate interface trap states by coordinating uncoordinated lead ions while covalently anchoring oxide hydroxyls [58,59]. Uncertified champion devices with L-isoleucine modification reach 26.15% PCE versus 23.54% for controls [58], while sulfonated bathocuproine derivatives suppress recombination through combined sodium-ion and pyridinic coordination to reach 26.02% PCE [59]. Decoupling bulk crystallization from interfacial reconstruction via triple-alkali interlayers provides an V_{oc} of 1.184 V and 83.81% fill factor (champion 26.13% PCE) [60], demonstrating that chemical restructuring of the contact outweighs electrostatic band alignment.

6. Operational Lifetime and Failure Pathways

Accelerated operational testing frequently fails to distinguish extrinsic contact corrosion, edge ingress, and phase segregation from intrinsic absorber degradation [61,62,63]. In triple-junction perovskite/perovskite/silicon cells, unencapsulated outdoor operation degrades efficiency to 80% after 4 months and to 50% after 6 months [63]. This diagnostic ambiguity is compounded by the scarcity of standardized lifetime evaluations: only 6 of 167 closely read papers reported a time to 80% of initial efficiency (T80) and only 3 named an ISOS protocol, such as ISOS-D-1 [64] or ISOS-D [65].

At electron-selective contacts, phenyl-C61-butyric acid methyl ester (PCBM) degrades via photo-induced [2+2] cycloaddition, causing dimerization and morphological instability that steric hindrance from nonplanar additives suppresses, reaching an uncertified champion PCE of 26.6% [66]. At heterointerfaces, bulky anionic passivators induce thermal lattice expansion, whereas reactive reconstruction and embedded passivation mitigate this fatigue, enabling unencapsulated devices to achieve a certified 27.28% efficiency with a 3566 h T80 and retaining 87% of initial efficiency after 4092 h of MPPT under ISOS-L-1 [4]. Degradation of self-assembled monolayers (SAMs) during perovskite deposition is mitigated by transient two-dimensional capping layers [67] or hydroxylated MXenes on nickel oxide [68], though both demonstrations remain restricted to sub-0.1 cm² apertures of 0.05 cm² [67] and 0.0524 cm² [68].

In the bulk absorber, deep-level interstitial defects dominate non-radiative degradation; electropositive ion passivation reduces rear-interface energy loss from 1.46% to 0.62% to yield a certified 25.56% efficiency [69]. Relieving lattice microstrain stabilizes mixed-cation absorbers, enabling unencapsulated 12 cm² devices to retain over 94.36% of initial efficiency after 1500 h under ISOS-L-1I and 80.5% after 1000 h at 85°C under ISOS-D-2I [10]. Conversely, ionic sulfur anchoring of lithium dopants preserves approximately 90% of initial efficiency after 800 h of MPPT, but this remains demonstrated only on an uncertified 0.04 cm² champion cell reaching 26.09% [70].

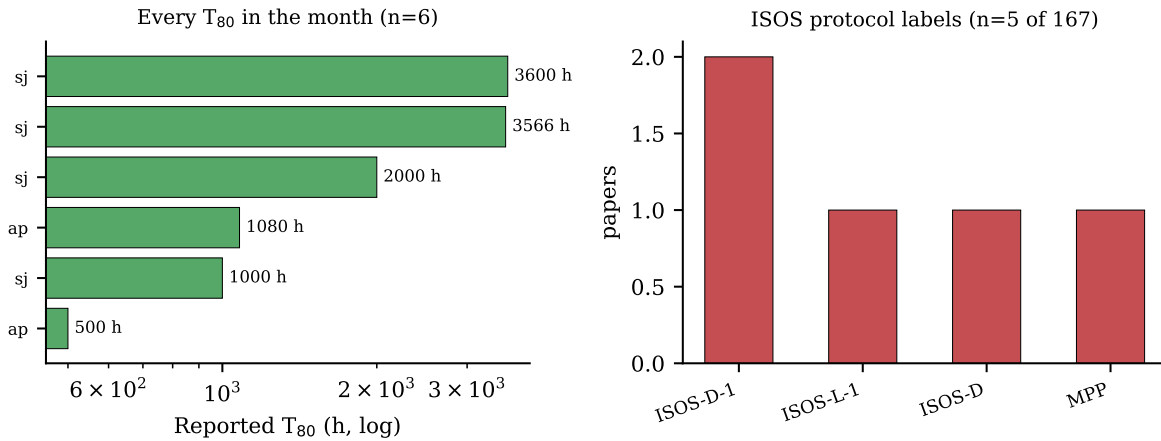


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

7. Subcell Design and Tandem Interconnection

Macroscopic current-voltage measurements cannot distinguish benign structural disorder from performance-limiting defects: five sequentially evaporated scaffolds yielded identical single-junction PCE of about 19% (supporting an uncertified 28.4% champion tandem PCE) despite marked disparities in conversion, halide distribution, and morphology [71]. Nor can standard diagnostics separate electronic losses from mechanical fatigue, as bending stress in flexible all-perovskite tandems (uncertified champion PCE of 24.97%) accelerates non-radiative recombination detectable by grazing-incidence X-ray diffraction (GIXRD) well before crack formation [72]. What survives this ambiguity is that targeted chemical passivation and redox regulation reliably mitigate recombination and ion migration across buried contacts [1,3,9]. Anchoring dense SAM coatings on reactive plasma deposition (RPD)-grown tin oxide (SnOx) suppresses halide migration in indium-free perovskite/silicon tandems, securing a certified PCE of 33.6% (uncertified champion 33.2% on 1 cm²) and certified 31.0% on a 207.9 cm² minimodule [1]. At hole contacts, neutralized alkali metal-based phosphonate salts (2PACz-M) mixed with Me-4PACz delocalize interfacial charge across 29.7 cm² modules (uncertified champion 26.88% PCE) [9], while embedding allylthiourea (ATU) into poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) enables dynamic iodine recycling, yielding a certified 28.79% PCE (uncertified champion 29.29%) [3]. In narrow-bandgap subcells, Lewis coordination and seeding templates converge to govern crystallite growth: Lewis hard-basicity amplification achieved an uncertified 23.71% single-junction PCE and an uncertified 29.44% champion tandem PCE with an uncertified time to 80% initial efficiency (T80) of 1080 h [73], while hybrid CsPbBr₃ quantum dot and metal-organic framework templates raise FF above 81% (uncertified 23.32% single-junction PCE; uncertified 29.18% tandem PCE, uncertified T80 of 500 h) [74]. For scale-up, these chemical formulations endure atmospheric and mechanical stress: blade coating with low-hygroscopic alcohol-alkane mixtures at 80% relative humidity achieves a certified 26.1% PCE [75], while dynamic-bond cross-linking monomers permit self-curing at 40°C for 30 min to deliver a certified 26.8% PCE (uncertified champion 27.12%) on 655.2 cm² flexible modules [6].

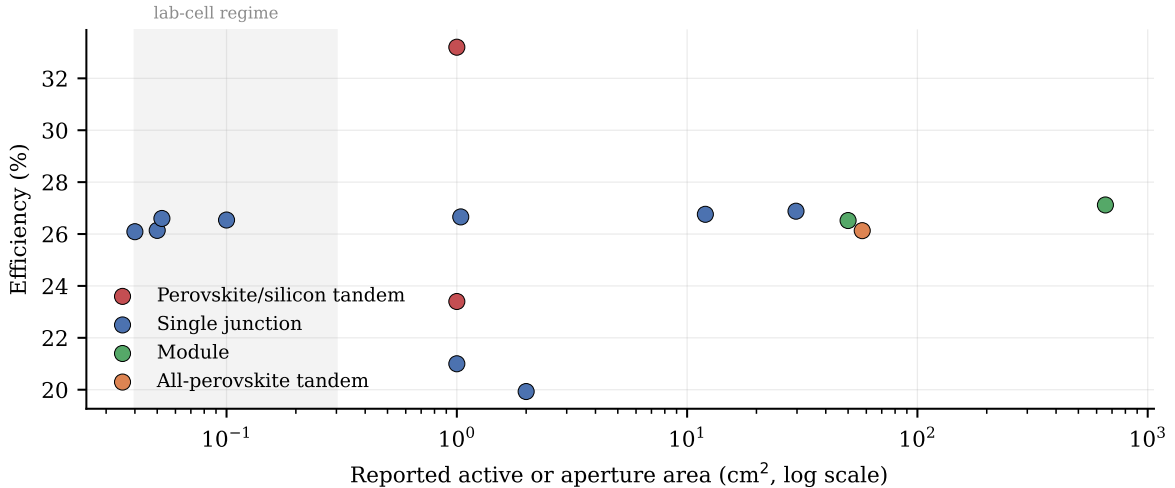


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

8. Research Gaps and Outlook

Standard characterization cannot distinguish electronic defect remediation from transient ionic polarization, electrostatic band-bending, or kinetic phase trapping. Certified PCE reached 33.6% in two-terminal (2T) tandems [1] and 26.8% across a 655.2 cm² aperture [6], yet only 14 of 167 closely read papers reported certified values, leaving most performance claims unresolved against measurement artifacts. Independent verification remains scarce across the literature, where the certified reporting share was 11.6 in 2026-05 and 5.9 in 2026-06.

First, standard optoelectronic characterization fails to establish whether Lewis-base chelators permanently remediate deep sub-gap traps or merely induce transient field-effect screening [12]. Differentiating true chemical de-trapping from dynamic ionic screening requires operando, bias-dependent drive-level capacitance profiling across hole transport layer (HTL) junctions, where interfacial trapped-hole densities reach $1.6 \times 10^{17} \text{ cm}^{-3}$ [48].

Second, bulk structural probes do not establish whether strain-regulating additives shift the equilibrium free-energy landscape or merely impose kinetic barriers that freeze metastable phases [32]. Proving whether interfacial coherent strain suppresses yellow non-perovskite phase nucleation requires in situ high-pressure synchrotron XRD paired with differential scanning calorimetry calibrated against absolute transition enthalpies [34].

Third, unstandardized operational metrics fail to establish the boundary separating reversible ionic polarization from catastrophic degradation under electrical stress [62]. Only 6 papers reported a time to 80% of initial efficiency (T80) lifetime (longest 3600 h) and only 3 named an ISOS protocol [4,61]. Closing this gap demands operando impedance spectroscopy tracking equivalent-circuit transport parameters during cyclic ISOS reverse-bias stress [4,62].

Fourth, macroscopic current-voltage scans cannot establish which morphological disparities in sequentially deposited scaffolds penalize carrier collection [71]. Although scale-up was addressed by 24 of 623 works, identifying performance-limiting defects over large areas requires spatially resolved photoconductive atomic force microscopy correlated with local photoluminescence quantum yields across device apertures up to 655.2 cm² [6].

This month’s evidence could not determine whether multidentate chelation or contact reconstruction provides greater operational durability than untreated junctions, because reported aging tests lacked common electrical bias and thermal baselines. Establishing that comparison requires reporting continuous degradation rates under matched ISOS protocols on designated aperture areas, tracking equivalent-circuit series resistance and impedance evolution rather than isolated champion retention points.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. The pipeline code, the stored run artifacts, the gate reports and the rendered documents for every issue are openly available at <https://github.com/havidaqoma/auto-perov-review-public>. Abstract text and full texts are not redistributed.

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